

Intera 0.5 T	Intera 1.0 T		Intera 1.5 T		
Standard	Omni	Power	Omni	Power	Master

Section 19**Physiology**

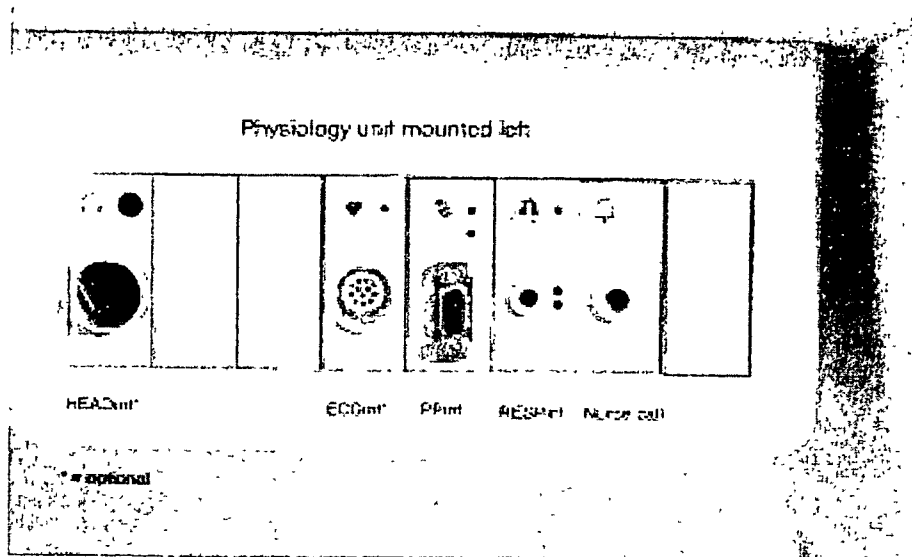
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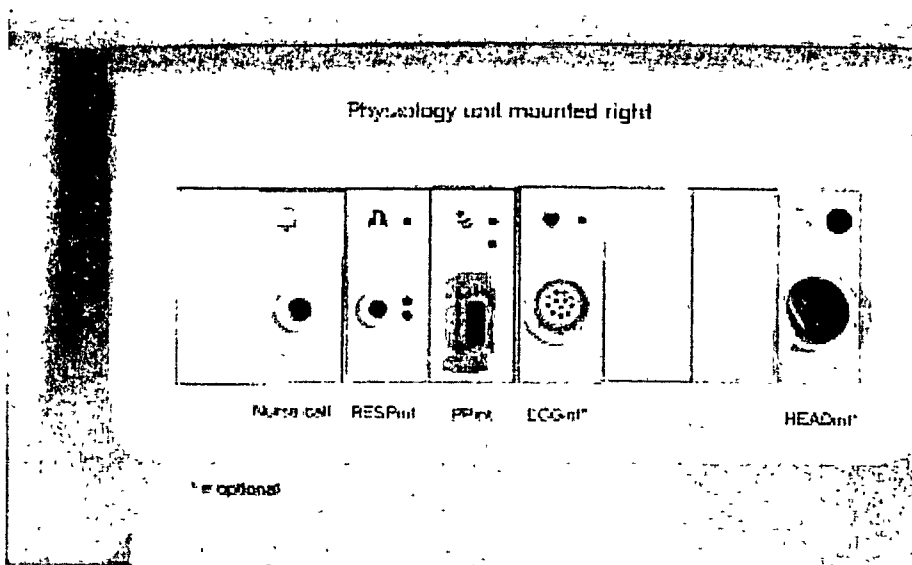
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1. PHYSIOLOGY CHAIN

The physiology modules of the Intera-system are installed in the PHYSIOLOGY unit, which communicates with the BDAS through the PFINT board of the PFEI unit and DFEC1.



Picture 1 Module arrangement when the physiology unit is mounted at left side.



Picture 2 Module arrangement when the physiology unit is mounted at right side.

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Inside the PHYSIOLOGY unit the following physiology modules are available:

ECG *	ECG module *
RESP	Respiratory module
PPint	Peripheral Pulse Module
NC	Nurse call

* = optional

The entire physiology chain can be checked with the automatic test programs in TBDAS. This functional automatic test has been done in a previous section.

2. PHYSIOLOGY UNIT FINAL TESTS

To check the proper functioning of the peripheral pulse unit (PPU) and respiratory gating (RESP the application software should be used.

1. Position the head performance phantom with the holder for transversal scans in the head coil. Connect the head coil to one of the SC-connectors. Put the head coil into the iso-centre of the magnet by pressing <Travel to scanplane>.

CAUTION

FIBER OPTIC SENSOR CABLE: Glass fiber sensor cable is fragile. The cable should never be flexed or bent at extreme angles smaller than 3 inch radius to avoid breaking fiber optic strands within the cable. Use proper care not to pinch or step on the cable or connectors.

2. Connect the 'pulse sensor' to a volunteer, who stands near the patient support.
It is advised to connect the 'pulse sensor' to the ring finger. Tell the volunteer to rest his arm with the sensor on the table top, because the automatic gain adjustment may cause a loss of trigger signal after a movement.
3. Login under GYROSCAN
4. Select: New Exam
Select: 'Dummy Patient' or create 'dummy patient'
Select: Anatomy
Select: Phantom studies
Select: HEADT
Select: H11:MS,SE,30
Select: Motion
5. Modify. Cardiac synchronization Trigger PPU
6. Select: <Proceed>
7. Select: PPU in the physiology window on your screen.
8. Perform the scan, select: <START SCAN>

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9. When the scan is finished:
10. Remove PPU sensor from the volunteer.
11. Connect the 'RESP sensor' to a volunteer, which stands near the patient support. It is advised to connect the 'RESP sensor' on the midriff, underneath the clothes. Tell the volunteer not to move, because the automatic gain adjustment may cause a loss of trigger signal after a movement.
12. Connect the 'RESP sensor' to the RESP module.
13. Select: Scan list
Select: <Proceed>
Select: Motion
14. Modify: Cardiac synchronization No
Respiratory compensation PEAR
15. Select: <Proceed>
16. Select: RESPIRATION in the physiology window on your screen
17. Perform the scan, select: <START SCAN>

3. PATIENT ALARM (NURSE CALL)

3.1. INTRODUCTION

The alarm is meant for patients, who in case of emergency, only can react by squeezing the pinch ball. If the patient squeezes the ball a buzzer is activated, to alarm the operator.

3.2. TESTING THE PATIENT ALARM

Procedure:

Test the Patient Alarm Box, by squeezing the pinch ball. One should hear two beeps with a slight interval. The first beep is the Patient Alarm Box (hard wired nurse call), the second beep is the sound out of the Audio Module (the software controlled nurse call).

Write down the test result in the MRL (section 22, Safety checks)